

Real Vector Space

A real vector space is a set of elements V together with two operations $\oplus$ and $\odot$ satisfying the following properties:

(α) If $u, v \in V$, then $u \oplus v \in V$

(i.e. V is closed under the $\oplus$ operation)

(a) $u \oplus v = v \oplus u$, for $u, v \in V$

(b) $u \oplus (v \oplus w) = (u \oplus v) \oplus w$, for $u, v, w \in V$

(c) There is an element 0 in V s.t.

$$u \oplus 0 = 0 \oplus u = u \quad \forall u \in V$$

(d) For each $u \in V$, $\exists -u \in V$ s.t.

$$u \oplus -u = 0$$

(β) If $u \in V$ and c is any real no.

then $c \odot u \in V$ (i.e. V is closed under the $\odot$ operation)

(e) $c \odot (u \oplus v) = c \odot u \oplus c \odot v$, $\forall$ real no c
and $u, v \in V$

(f) $(c + d) \odot u = c \odot u \oplus d \odot u$, $\forall$ real no $c \neq d$
and $u \in V$

(g) $c \odot (d \odot u) = (cd) \odot u$, $\forall$ real no $c \neq d$
and $u \in V$

(h) $1 \odot u = u \quad \forall u \in V$

Ex . Set of real nos. in $\mathbb{R}^n$ space

$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_i \in \mathbb{R}\}$ is a vector space with usual addition and scalar multiplication

Pr

Let $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n)$

$\& z = (z_1, z_2, \dots, z_n) \in \mathbb{R}^n$

Sum of any two real nos. in $\mathbb{R}^n$ space again in $\mathbb{R}^n$

$$\text{ie } x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) \in \mathbb{R}^n$$

$\Rightarrow \mathbb{R}^n$ is closed under addition.

$$\begin{aligned} \textcircled{1} \quad x + y &= (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) \\ &= (y_1 + x_1, y_2 + x_2, \dots, y_n + x_n) \\ &= y + x \quad \therefore \text{Commutative} \end{aligned}$$

$$\textcircled{2} \quad x + (y + z) = (x + y) + z \quad \text{is also true (Associativity)}$$

$$\textcircled{3} \quad \therefore (0, 0, \dots, 0) \in \mathbb{R}^n \Rightarrow \text{zero element in } \mathbb{R}^n$$

$$\textcircled{4} \quad (-x_1, -x_2, \dots, -x_n) \text{ is inverse of } (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$$

$\Rightarrow (\mathbb{R}^n, +)$ is an abelian group.

$$\textcircled{5} \quad \text{Let } x = (x_1, x_2, \dots, x_n) \& y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n \\ \text{and } \alpha, \beta \in \mathbb{R}$$

$$\begin{aligned} \alpha(x + y) &= \alpha[(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n)] \\ &= \alpha(x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) \\ &= (\alpha x_1 + \alpha y_1, \alpha x_2 + \alpha y_2, \dots, \alpha x_n + \alpha y_n) \\ &= \alpha(x_1, x_2, \dots, x_n) + \alpha(y_1, y_2, \dots, y_n) \\ &= \alpha x + \alpha y \end{aligned}$$

$$(6) (\alpha + \beta)x = \alpha x + \beta x$$

$$\begin{aligned} (\alpha + \beta)(x_1, x_2, \dots, x_n) &= ((\alpha + \beta)x_1, (\alpha + \beta)x_2, \dots, (\alpha + \beta)x_n) \\ &= (\alpha x_1, \alpha x_2, \dots, \alpha x_n) + (\beta x_1, \beta x_2, \dots, \beta x_n) \\ &= \alpha x + \beta x \end{aligned}$$

$$(7) \alpha(\beta x) = (\alpha\beta)x$$

$$\begin{aligned} \alpha(\beta(x_1, x_2, \dots, x_n)) &= (\alpha\beta)(x_1, x_2, \dots, x_n) \\ &= (\alpha\beta)x \end{aligned}$$

$$(8) 1 \cdot x = 1 \cdot (x_1, x_2, \dots, x_n) = (x_1, x_2, \dots, x_n) = x$$

$\therefore \mathbb{R}^n$ is a vector space over $\mathbb{R}$.

Ex

The set $M_2(\mathbb{R})$ of all 2×2 matrices is a vector space over $\mathbb{R}$ under matrix addition & scalar multiplication defined by

$$\alpha \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \alpha a & \alpha b \\ \alpha c & \alpha d \end{pmatrix}$$

Ex

$P_n = \{a_0 + a_1x + \dots + a_nx^n : a_i \in F\}$ with the usual polynomial addition & scalar multiplication is a vector space over F .

Ex

Let V be the set of all real numbers with operations $u \oplus v = u - v$

and $c \odot u = cu$.

Is V a vector space?

Soln

Let $u, v \in V$ & c is a scalar

$$u \oplus v \in V \text{ \& } c \odot u \in V$$

$\Rightarrow$ closed under $\oplus$ & $\odot$ sign.

Commutativity

$$u \oplus v = v \oplus u$$

Let $u = 2$ & $v = 3$

$$u \oplus v = 2 - 3 = -1$$

$$v \oplus u = 3 - 2 = 1$$

$\Rightarrow$ Not commutative

Also Associativity, additive identity & additive inverse do fail to hold

$$\left. \begin{aligned} c \odot (u \oplus v) &= c \odot u \oplus c \odot v \\ c \odot (d \odot u) &= (cd) \odot u \\ 1 \odot u &= u \end{aligned} \right\} \text{ holds}$$

Fail on $(c+d) \odot u = c \odot u \oplus d \odot u$

L.H.S. $\Rightarrow (c+d) \odot u = (2+3) \odot 4 = 5 \odot 4 = 20$

R.H.S. $\Rightarrow c \odot u \oplus d \odot u = 2 \odot 4 \oplus 3 \odot 4 = 8 \oplus 12 = -4$

$\Rightarrow V$ is not a vector space.

Ex Let $V = \mathbb{R}^2$

$$[x, y] \oplus [w, z] = [x+w+1, y+z-2]$$

$$a \odot [x, y] = [ax+a-1, ay-2a+2]$$

Form a vector space? Yes

Solⁿ

closure property satisfies since resultant vector w.r.t.

$\oplus$ & $\odot$ is in $\mathbb{R}^2$ space

(1) $(x, y) \oplus (u, v) = (u, v) \oplus (x, y)$ holds

(2) Associativity

$$(x, y) \oplus [(u, v) \oplus (w, z)] = [(x, y) \oplus (u, v)] \oplus (w, z)$$

L.H.S. $(x, y) \oplus (u+w+1, v+z-2)$
 $= (x+u+w+2, y+v+z-4)$

R.H.S. $(x+u+1, y+v-2) \oplus (w, z)$
 $= (x+u+w+2, y+v+z-4)$

(3) Additive identity

$$\text{let } (x, y) \in \mathbb{R}^2$$

$$(x, y) \oplus (0, 0) = (x+0+1, y+0-2)$$

$$\Rightarrow x = -1, y = 2$$

$\Rightarrow (-1, 2)$ will work as zero vector

(4) Additive inverse

$$(x, y) \in \mathbb{R}^2 \quad \exists (u, v) \text{ is inverse element}$$

$$(x, y) \oplus (u, v) = (-1, 2)$$

$$(x+u+1, y+v-2) = (-1, 2)$$

$$u = -x-2, \quad v = -y+4$$

$\Rightarrow (-x-2, -y+4)$ is the additive inverse

(5) $c \odot [(x, y) \oplus (w, z)] = c \odot (x, y) \oplus c \odot (w, z)$

L.H.S. $= c \odot (x+w+1, y+z-2)$

$$= (cx + cw + 2c - 1, cy + cz - 2c - 2c + 2)$$

$$= (cx + cw + 2c - 1, cy + cz - 4c + 2)$$

$$R.H.S = c \odot (x, y) \oplus c \odot (w, z)$$

$$= (cx + c - 1, cy - 2c + 2) \oplus (cw + c - 1, cz - 2c + 2)$$

$$= (cx + c - 1 + cw + c - 1 + 1, cy + cz - 4c + 4 - 2)$$

$$= (cx + cw + 2c - 1, cy + cz - 4c + 2)$$

$$(6) (c+d) \odot (x, y) = c \odot (x, y) \oplus d \odot (x, y)$$

$$(7) (cd) \odot (x, y) = c \odot [d \odot (x, y)]$$

$$\underline{L.H.S.} \quad (cd) \odot (x, y) = [cdx + cd - 1, cdy - 2cd + 2]$$

$$\underline{R.H.S.} \quad c \odot [d \odot (x, y)] = c \odot [dx + d - 1, dy - 2d + 2]$$

$$= (cdx + cd - c + c - 1, cdy - 2cd + 2c - 2c + 2)$$

$$= (cdx + cd - 1, cdy - 2cd + 2)$$

$$(8) 1 \odot (x, y) = (x, y)$$

$$\underline{L.H.S} \quad (x + 1 - 1, y - 2 + 2) = (x, y) = \underline{R.H.S.}$$

Q Let ~~V~~ be the set of all ^{ve} real numbers

Q The set of all ^{ve} real no. u with the operations $u \oplus v = uv$ and $c \odot u = u^c$ Vector space ?

Thm Let V be a vector space and let x, y be vectors in V . Then

- ① $x + y = y \Rightarrow x = 0$,
- ② $0x = 0$
- ③ $k0 = 0$ for any $k \in \mathbb{R}$
- ④ $-x$ is unique and $-x = (-1)x$
- ⑤ If $kx = 0$, then $k = 0$ or $x = 0$

Proof $\Rightarrow$ ① Adding $-y$ to both sides of $x + y = y$

$$x = x + 0 = x + y + (-y) = y + (-y) = 0$$

② $0x = (0+0)x = 0x + 0x \Rightarrow 0x = 0$ by ①

④ The uniqueness of the $-ve$ $-x$ of x can be shown by simple modification in thm.

if $\bar{x}$ is another negative of x s.t. $x + \bar{x} = 0$,

$$\begin{aligned} \text{then } -x &= -x + 0 = -x + (x + \bar{x}) = (-x + x) + \bar{x} \\ &= 0 + \bar{x} = \bar{x} \end{aligned}$$

on the other hand, the eqn

$$x + (-1)x = 1x + (-1)x = (1-1)x = 0x = 0$$

shows that $(-1)x$ is another $-ve$ of x , and hence

$-x = (-1)x$ by the uniqueness of $-x$.

⑤ Suppose $kx = 0$ and $k \neq 0$

$$\text{then } x = 1x = \frac{1}{k}(kx) = \frac{1}{k} \times 0 = 0$$

Thm A nonempty subset W of a vector space V is a subspace if and only if $x+y$ and kx are contained in W (or equivalently, $x+ky \in W$) for any vectors x and y in W and any scalar $k \in \mathbb{R}$.

Subspaces

Let V be a vector space and W a nonempty subset of V . If W is a vector space w.r.t. the operations in V , then W is called a subspace of V .

- * Every vector space has at least two subspaces. (i) $\{0\}$ the subspace $\{0\}$ consisting of only of the zero vector.

$$[0 \oplus 0 = 0 \text{ \& } c \odot 0 = 0 \text{ in any vector space}]$$

The subspace $\{0\}$ is called the zero subspace

Ex

Let $W \subseteq \mathbb{R}^3$ consist vector of the form $(a, b, 0)$,

$$\text{Let } u = (a, b, 0) \text{ \& } v = (c, d, 0)$$

$$\text{then } u + v = (a + c, b + d, 0) \in W$$

c any scalar

$$cu = c(a, b, 0) = (ca, cb, 0) \in W$$

$\Rightarrow$ closure property holds.

Similarly other properties also hold $\Rightarrow W$ is subspace of $\mathbb{R}^3$.

Thm

Let V be a vector space with operations $\oplus$ and $\odot$ and let W be a nonempty subset of V . Then W is a subspace of V iff .

(a) If $u, v \in W$, then $u \oplus v \in W$

(b) If c any scalar \& $u \in W$ then $c \odot u \in W$

- * The subspace consisting only of the zero vector is a nonempty subspace.

- * A vector space V itself and the zero vector $\{0\}$ are trivially subspaces.

Ex

Let W be the set of all 2×3 matrices of form
$$\begin{bmatrix} a & b & 0 \\ 0 & c & d \end{bmatrix}$$
 where a, b, c & d are real no.

Let $u = \begin{bmatrix} a & b & 0 \\ 0 & c & d \end{bmatrix}$ & $v = \begin{bmatrix} e & f & 0 \\ 0 & g & h \end{bmatrix} \in W$

$$u+v = \begin{bmatrix} a+e & b+f & 0 \\ 0 & c+g & d+h \end{bmatrix} \in W$$

if k is any scalar

$$ku = \begin{bmatrix} ka & kb & 0 \\ 0 & kc & kd \end{bmatrix} \in W$$

$\Rightarrow W$ is a subspace of M_{23} .

* A non-empty subset W of a vector space V is a subspace of V iff $au+bv$ is in W for any $u, v \in W$ and any scalars a and b .

or $u+av$

or $au+av \in W$

Ex

Let W be the set of points inside and on the unit circle in the xy -plane

$W = \{(x, y) : x^2 + y^2 \leq 1\}$ then verify whether W is a subspace or not of $\mathbb{R}^2$

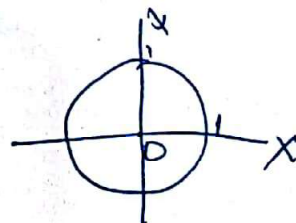
Soln

Given $W = \{(x, y) : x^2 + y^2 \leq 1\}$

Let $u = (\frac{1}{2}, \frac{1}{2}) \in W$ for $\alpha = 4 \in \mathbb{R}$

$\alpha u = (2, 2) \notin W$

$\Rightarrow W$ is not a subspace of $\mathbb{R}^2$.



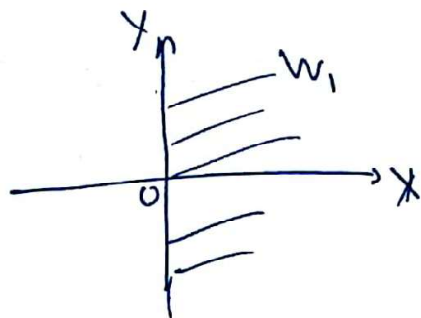
Ex

check whether subspace of $\mathbb{R}^2$ or not

(a) W_1 is the set of all vectors of the form $\begin{bmatrix} x \\ y \end{bmatrix}$

where $x \geq 0$

Soln



$\therefore W$ is right half of xy -plane
It is not subspace of $\mathbb{R}^2$

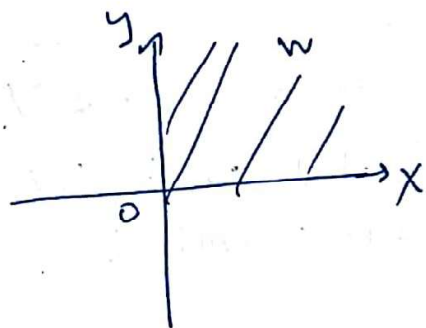
$$\therefore \text{if } \begin{bmatrix} 2 \\ 3 \end{bmatrix} \in W,$$

$$\text{Let } \alpha = -3$$

$$-3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -6 \\ -9 \end{bmatrix} \notin W,$$

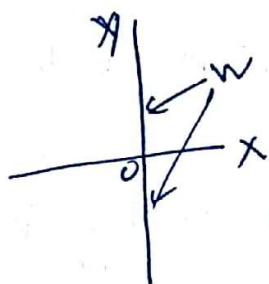
$\Rightarrow W_1$ is not subspace

(b) W is set of all vectors of the form $\begin{bmatrix} x \\ y \end{bmatrix}$, $x \geq 0$, $y \geq 0$.



Not a subspace of $\mathbb{R}^2$

(c) W is set of all vectors of the form $\begin{bmatrix} x \\ y \end{bmatrix}$, where $x = 0$



$$\text{Let } u = \begin{bmatrix} 0 \\ b \end{bmatrix}, v = \begin{bmatrix} 0 \\ c \end{bmatrix}$$

$$u + v = \begin{bmatrix} 0 \\ b \end{bmatrix} + \begin{bmatrix} 0 \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ b+c \end{bmatrix} \in W$$

$$cu = c \begin{bmatrix} 0 \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ cb \end{bmatrix} \in W$$

$\Rightarrow W$ is subspace of $\mathbb{R}^2$.

Ex

W is subset of $\mathbb{R}^3$ consisting vectors of the form $(a, b, 1)$.

→ Not a subspace.

Ex

Let W be the set of vectors in $\mathbb{R}^3$ of the form $[a, b, \frac{1}{2}a - 2b]$, is W subspace of $\mathbb{R}^3$

$$[a_1, b_1, \frac{1}{2}a_1 - 2b_1] + [a_2, b_2, \frac{1}{2}a_2 - 2b_2]$$

$$= [a_1 + a_2, b_1 + b_2, \frac{1}{2}a_1 - 2b_1 + \frac{1}{2}a_2 - 2b_2]$$

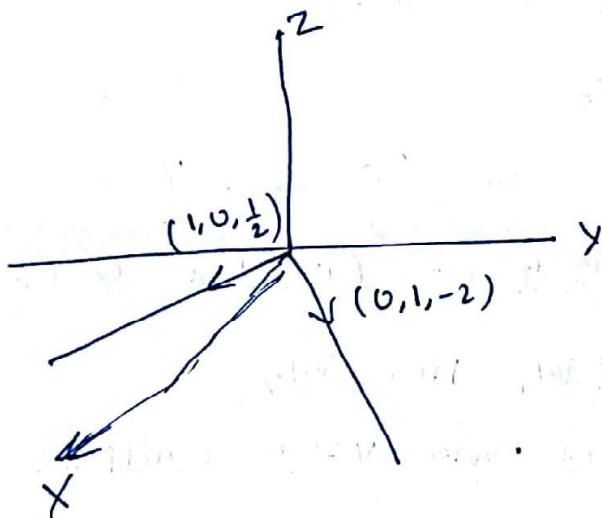
$$= [a_1 + a_2, b_1 + b_2, \frac{1}{2}(a_1 + a_2) - 2(b_1 + b_2)]$$

$$k[a, b, \frac{1}{2}a - 2b] = [ka, kb, k(\frac{1}{2}a - 2b)]$$

$$= [ka, kb, \frac{1}{2}(ka) - 2(kb)]$$

$$* [a, b, \frac{1}{2}a - 2b] = a[1, 0, \frac{1}{2}] + b[0, 1, -2]$$

W consists of the set of all linear combinations of $[1, 0, \frac{1}{2}]$ and $[0, 1, -2]$



Geometrically, W is the plane in $\mathbb{R}^3$ through origin containing the vectors

$$[1, 0, \frac{1}{2}], [0, 1, -2].$$

This plane is the set of all possible destinations using these two directions.

* Arbitrary intersection of subspaces of V is a subspace of V .

Prp Let $\{W_i\}$ be the collection of all possible subspaces of V & let $W = \bigcap_i W_i$

$\therefore$ each W_i is a subspace of V

$$\Rightarrow 0 \in W_i \quad \forall i$$

$$\Rightarrow 0 \in \bigcap_i W_i = W$$

$$\text{Let } u, v \in W = \bigcap_i W_i$$

$$\Rightarrow u, v \in W_i \quad \forall i$$

$$\alpha u + v \text{ or } u + \alpha v \in W_i$$

$$\Rightarrow \alpha + uv \in \bigcap_i W_i = W$$

$\therefore W$ is a subspace of V .

* Union of two subspaces need not be a subspace

$$\text{let } W_1 = \{(0, 0, x) : x \in \mathbb{R}\}$$

$$\& W_2 = \{(0, y, 0) : y \in \mathbb{R}\}$$

$$\text{let } W = W_1 \cup W_2$$

$$\therefore (0, 0, x) \in W \text{ and } (0, y, 0) \in W$$

$$\text{but } (0, 0, x) + (0, y, 0) = (0, y, x) \notin W$$

$$\text{neither } (0, y, x) \in W_1 \text{ nor } W_2$$

$$\Rightarrow W \text{ is not closed under vector addition}$$

$$\therefore \text{not a subspace}$$

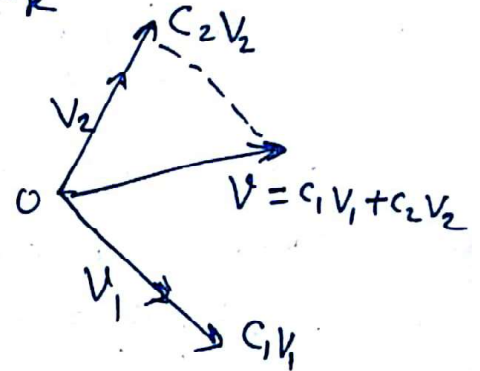
Linear Combination

Let $v_1, v_2, \dots, v_k$ be vectors in a vector space V .

A vector v in V is called a linear combination of $v_1, v_2, \dots, v_k$ if

$$v = c_1 v_1 + c_2 v_2 + \dots + c_k v_k$$

for some real numbers $c_1, c_2, \dots, c_k$



Ex

Let $v_1 = (1, 2, 1)$, $v_2 = (1, 0, 2)$ and $v_3 = (1, 1, 0)$ in $\mathbb{R}^3$

$v = (2, 1, 5)$ is a linear combination of v_1, v_2, v_3 if we can find real numbers c_1, c_2 and c_3 so that

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = v$$

$$c_1 (1, 2, 1) + c_2 (1, 0, 2) + c_3 (1, 1, 0) = (2, 1, 5)$$

$$c_1 + c_2 + c_3 = 2$$

$$2c_1 + c_3 = 1$$

$$c_1 + 2c_2 = 5$$

$$\Rightarrow c_1 = 1, \quad c_2 = 2, \quad c_3 = -1$$

$$\Rightarrow v = v_1 + 2v_2 - v_3$$

$\Rightarrow v$ is a l.c. of $v_1, v_2, \Delta v_3$

Linear Combinations Span S

g) $S = \{v_1, v_2, \dots, v_k\}$ is a set of vectors in V , then the set of all vectors in V that are linear combination of the vectors in S is denoted by $\text{Span } S$ or $\text{Span}\{v_1, v_2, \dots, v_k\}$

Ex $S = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$

Then $\text{Span } S$ is the set in M_{23} consisting of all vectors of the form

$$a \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} a & b & 0 \\ 0 & c & d \end{bmatrix}, \text{ where } a, b, c \text{ \& } d \text{ are real n.}$$

i.e. $\text{Span } S$ is the subset of M_{23} consisting of all matrices of the form $\begin{bmatrix} a & b & 0 \\ 0 & c & d \end{bmatrix}$

Th^m Let $S = \{v_1, v_2, \dots, v_k\}$ be a set of vectors in a vector space V . Then $\text{Span } S$ is a subspace of V .

Ex In P_2 let $v_1 = 2t^2 + t + 2$, $v_2 = t^2 - 2t$, $v_3 = 5t^2 - 5t + 2$
 $v_4 = -t^2 - 3t - 2$

Determine if the vector $u = t^2 + t + 2$ belongs to $\text{Span}\{v_1, v_2, v_3, v_4\}$.

Linear Independence

Defn. The vectors $v_1, v_2, \dots, v_k$ in a vector space V are said to span V if every vector in V is a linear combination of $v_1, v_2, \dots, v_k$.

if $S = \{v_1, v_2, \dots, v_k\}$, then we also say that the set S spans V , or that $\{v_1, v_2, \dots, v_k\}$ spans V , or V is spanned by S , or $\text{Span } S = V$.

Procedure to check if vectors $v_1, v_2, \dots, v_k$ span the vector space V

Step I - Choose an arbitrary vector v in V .

Step II - Determine if v is a linear combination of the given vectors. If it is, then the given vectors span V . If it is not, they do not span V .

Ex Let V be the vector space $\mathbb{R}^3$ and let

$$v_1 = (1, 2, 1) \quad v_2 = (1, 0, 2) \quad \text{and} \quad v_3 = (1, 1, 0)$$

Do v_1, v_2, v_3 span V ?

Soln Let $v = (a, b, c) \in \mathbb{R}^3$ where a, b & c are arbitrary const

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = v$$

$$c_1 + c_2 + c_3 = a$$

$$2c_1 + c_3 = b$$

$$c_1 + 2c_2 = c$$

$$\text{A soln } c_1 = \frac{-2a+2b+c}{3}, \quad c_2 = \frac{a-b+c}{3}, \quad c_3 = \frac{4a-b-2c}{3}$$

$\Rightarrow v_1, v_2, v_3$ span $\mathbb{R}^3$.

Ex

Let V be the vector space P_2 .

$$\text{Let } S = \{p_1(t), p_2(t)\}$$

$$\text{where } p_1(t) = t^2 + 2t + 1$$

$$\text{and } p_2(t) = t^2 + 2$$

Does S span P_2 ?

Solⁿ

Let $p(t) = at^2 + bt + c$ be any polynomial in P_2

$$p(t) = c_1(t^2 + 2t + 1) + c_2(t^2 + 2)$$

$$\Rightarrow c_1 + c_2 = a$$

$$2c_1 = b$$

$$c_1 + 2c_2 = c$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & | & 2a-c \\ 0 & 1 & | & c-a \\ 0 & 0 & | & b-4a+2c \end{bmatrix}$$

$$\text{If } b-4a+2c \neq 0$$

$\Rightarrow$ System is inconsistent and there is no solⁿ

Hence $S = \{p_1(t), p_2(t)\}$ does not span P_2 .

[the polynomial $3t^2 + 2t - 1$ cannot be written as a linear combination of $p_1(t)$ and $p_2(t)$.]

Ex

The homogeneous linear system $Ax=0$, where

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 \\ -2 & -2 & 1 & -5 \\ 4 & 1 & -1 & 3 \\ 4 & 4 & -1 & 9 \end{bmatrix}$$

To determine the spanning set of this homogeneous system find the RREF form of matrix

The RREF

$$\begin{bmatrix} 1 & 1 & 0 & 2 & | & 0 \\ 0 & 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Gen. Solⁿ

$$x_1 = -x_2 - 2x_4$$

$$x_2 = x_4$$

$$x_3 = 0$$

$$x_4 = s$$

$$x = x_4 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} + s \begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Hence the vectors $\begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ span the solⁿ space

* The vectors $V_1, V_2, \dots, V_k$ in a vector space V are said to be linearly dependent if there exist constants $c_1, c_2, \dots, c_k$ not all zero

$$\text{s.t. } c_1 V_1 + c_2 V_2 + \dots + c_k V_k = 0$$

Otherwise $V_1, V_2, \dots, V_k$ are called linearly independent

$$\text{i.e. } c_1 = c_2 = \dots = c_k = 0$$

Steps to check LI or LD

I - Form the eqn $c_1 V_1 + c_2 V_2 + \dots + c_k V_k = 0$

II - In the homogeneous system obtained in I
has only the trivial soln, then LI,
If has nontrivial soln, then LD.

Ex. Are the vectors $V_1 = (1, 0, 1, 2)$, $V_2 = (0, 1, 1, 2)$
and $V_3 = (1, 1, 1, 3)$ in $\mathbb{R}^4$ LD or LI?

Soln $c_1 V_1 + c_2 V_2 + c_3 V_3 = 0$

$$c_1 + c_3 = 0$$

$$c_2 + c_3 = 0$$

$$c_1 + c_2 + c_3 = 0$$

$$2c_1 + 2c_2 + 3c_3 = 0$$

$$c_1 = c_2 = c_3 = 0$$

$\Rightarrow$ L.I.

Ex $V_1 = (1, 2, -1)$, $V_2 = (1, -2, 1)$, $V_3 = (-3, 2, -1)$
 $V_4 = (2, 0, 0)$ in $\mathbb{R}^3$. Is $S = \{V_1, V_2, V_3, V_4\}$ LD or LI

Sol Here non-trivial soln
 $\Rightarrow$ LD.

* The given vectors are LI iff $\det(A) \neq 0$

Q2

$$P_1(t) = t^2 + t + 2,$$

$$P_2(t) = 2t^2 + t$$

$$P_3(t) = 3t^2 + 2t + 2$$

$$c_1 P_1(t) + c_2 P_2(t) + c_3 P_3(t) = 0$$

$$c_1 + 2c_2 + 3c_3 = 0$$

$$c_1 + c_2 + 2c_3 = 0$$

$$2c_1 + 2c_3 = 0$$

$\Rightarrow$ has infinitely many solⁿ

Hence L.D.

Example (1) Do the polynomials $t^3 + 2t + 1$, $t^2 - t + 2$, $t^3 + 2$, $-t^3 + t^2 - 5t + 2$ span P_3 ?

(No)

(2) Find a set of vectors spanning the null space of

$$A = \begin{bmatrix} 1 & 1 & 2 & -1 \\ 2 & 3 & 6 & -2 \\ -2 & 1 & 2 & 2 \\ 0 & -2 & -4 & 0 \end{bmatrix}$$

$$\text{Ans } \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

(3) $x_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 1 \\ 6 \\ 3 \\ 0 \end{bmatrix}$ belong to the null space of A.

$\Rightarrow \{x_1, x_2, x_3\}$ L.I.?

(Yes)

Thm (3.4)

Let $v_1, v_2, \dots, v_m$ be vectors in a vector space V . Then the set $W = \{a_1 v_1 + a_2 v_2 + \dots + a_m v_m \mid a_i \in \mathbb{R}\}$

of all linear combinations of $v_1, v_2, \dots, v_m$ is a subspace of V called the subspace of V spanned by $v_1, v_2, \dots, v_m$.

Proof $\Rightarrow$

If W is subspace of V then it should be closed under addition and scalar multiplication.

Let $u, v \in W$

$$u = a_1 v_1 + a_2 v_2 + \dots + a_m v_m$$

$$v = b_1 v_1 + b_2 v_2 + \dots + b_m v_m$$

where a_i and b_i are scalars

$$u + v = (a_1 + b_1) v_1 + (a_2 + b_2) v_2 + \dots + (a_m + b_m) v_m$$

and for any scalar k ,

$$ku = (ka_1) v_1 + (ka_2) v_2 + \dots + (ka_m) v_m$$

$\therefore u + v$ and ku are linear combinations of $v_1, v_2, \dots, v_m$ and belong to W

$\Rightarrow W$ is a subspace of V .

Thm (3.5)

The nonzero rows of a matrix of a row-echelon form are linearly independent, and so are the columns that contain leading 1's.

Lemma (3.4) (1) Any set of n vectors in the m -space $\mathbb{R}^m$ is linearly dependent if $n > m$.

(2) If U is the RREF of A , then the columns of U are linearly independent iff the columns of A are linearly independent.

Basis

The vectors $v_1, v_2, \dots, v_k$ in vector space V are said to form a basis for V if

- (a) $v_1, v_2, \dots, v_k$ span V
- (b) $v_1, v_2, \dots, v_k$ are linearly independent

* The vectors $e_1, e_2, \dots, e_n$ form a basis for $\mathbb{R}^n$. Each of these sets of vectors is called the natural basis or standard basis for $\mathbb{R}^2, \mathbb{R}^3$, and $\mathbb{R}^n$.

Ex Show that the set $S = \{v_1, v_2, v_3, v_4\}$,
 $v_1 = (1, 0, 1, 0)$, $v_2 = (0, 1, -1, 2)$, $v_3 = (0, 2, 2, 1)$
 $v_4 = (1, 0, 0, 1)$ is a basis for $\mathbb{R}^4$.

Sol For L.I.

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 = 0$$

$$c_1 + c_4 = 0$$

$$c_2 + 2c_3 = 0$$

$$c_1 - c_2 + 2c_3 = 0$$

$$2c_2 + c_3 + c_4 = 0$$

$$c_1 = c_2 = c_3 = c_4 = 0$$

$\Rightarrow S$ is L.I.

For S span $\mathbb{R}^4$.

$$\text{Let } v = (a, b, c, d) \in \mathbb{R}^4$$

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 = v$$

S spans $\mathbb{R}^4$.

$\Rightarrow S$ is a basis for $\mathbb{R}^4$.

Ex Show that the set $S = \{t^2+1, t-1, 2t+2\}$ is a basis for the vector space P_2 .

Soln S must span V and L.I.

$$\text{Let } at^2+bt+c \in P_2$$

$$\begin{aligned} at^2+bt+c &= c_1(t^2+1) + c_2(t-1) + c_3(2t+2) \\ &= c_1t^2 + c_2t + 2c_3t + c_1 - c_2 + 2c_3 \end{aligned}$$

$$c_1 = a$$

$$c_2 + 2c_3 = b$$

$$c_1 - c_2 + 2c_3 = c$$

$$c_1 = a, \quad c_2 = \frac{a+b-c}{2}, \quad c_3 = \frac{c+b-a}{4}$$

Hence S spans V .

For L.I.

$$c_1(t^2+1) + c_2(t-1) + c_3(2t+2) = 0$$

$$c_1 = 0$$

$$c_2 + 2c_3 = 0$$

$$c_1 - c_2 + 2c_3 = 0$$

This homogeneous system has trivial soln only

$$\text{i.e. } c_1 = c_2 = c_3 = 0$$

$\Rightarrow$ L.I.

$\therefore S$ is a basis for P_2 .

* The set of vectors $\{t^n, t^{n-1}, \dots, t, 1\}$ forms a basis for the vector space P_n called the natural or standard basis for P_n .

* A Vector Space V is called finite-dimensional if there is a finite subset of V that is a basis for V .

eg. there is no such finite subset of V , then V is called infinite-dimensional. (Ex. $P \rightarrow$ Set of all polynomials $C(-\infty, \infty)$ all continuous function)

Thm (3.7) Let $\alpha = \{v_1, v_2, \dots, v_n\}$ be a basis for a vector space V . Then each vector x in V can be uniquely expressed as a linear combination of $v_1, v_2, \dots, v_n$, i.e., there are unique scalars c_i 's, $i=1, \dots, n$ s.t.

$$x = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

* The column vector $[c_1, c_2, \dots, c_n]^T$ is called the coordinate vector of x w.r.t. the basis α , and it is denoted by $[x]_\alpha$.

Q Determine if the set of vectors are bases for P_3

① $\{t^3 + 2t^2 + 3t, 2t^3 + 1, 6t^3 + 8t^2 + 6t + 4, t^3 + 2t^2 + t + 1\} \rightarrow \text{no}$

② $\{t^3 + t^2 + t + 1, t^3 + 2t^2 + t + 3, 2t^3 + t^2 + 3t + 2, t^3 + t^2 + 2t + 2\} \rightarrow \text{Forms basis for } P_3$

Proof Theorem (3.7) Let $v \in V$, $\Rightarrow v$ can be expressed as linear combination of elements of α , $\because \alpha$ is a basis.

$$v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n \quad \text{--- (1)}$$

$$\text{let } v = d_1 v_1 + d_2 v_2 + \dots + d_n v_n \quad \text{--- (2)}$$

$$\text{Eqn (1) - (2)} \Rightarrow 0 = (c_1 - d_1)v_1 + (c_2 - d_2)v_2 + \dots + (c_n - d_n)v_n$$

$\therefore$ Let the elements of α are L.I.

$$\Rightarrow c_1 - d_1 = 0, c_2 - d_2 = 0, \dots, c_n - d_n = 0$$

$$\Rightarrow c_i = d_i, 1 \leq i \leq n.$$

$\Rightarrow$ There is a unique way to express an element of V .

Dimension

The dimension of a non-zero vector space V is the no. of vectors in a basis of V .

$\dim V = n$ if there are n vectors in V .

Lemma 3.8 Let V be a vector space and let $\alpha = \{x_1, x_2, \dots, x_m\}$ be a set of m -vectors in V .

- (1) If α spans V , then every set of vectors with more than m vectors cannot be L.I.
- (2) If α is L.I., then any set of vectors with fewer than m vectors cannot span V .

Proof (2) follows (1).

For (1)

Let $\beta = \{y_1, y_2, \dots, y_n\}$ be a set of n -vectors in V with $n > m$.

$\therefore \alpha$ spans V

$\Rightarrow$ Every vector y_j is a linear combination of the vectors in the spanning set α .

$$y_j = a_{1j}x_1 + a_{2j}x_2 + \dots + a_{mj}x_m = \sum_{i=1}^m a_{ij}x_i$$

we have

$$\begin{aligned} c_1 y_1 + c_2 y_2 + \dots + c_n y_n &= c_1 (a_{11}x_1 + a_{21}x_2 + \dots + a_{m1}x_m) \\ &\quad + c_2 (a_{12}x_1 + a_{22}x_2 + \dots + a_{m2}x_m) \\ &\quad + \vdots \\ &\quad + c_n (a_{1n}x_1 + a_{2n}x_2 + \dots + a_{mn}x_m) \\ &= (a_{11}c_1 + a_{12}c_2 + \dots + a_{1n}c_n)x_1 + \\ &\quad (a_{21}c_1 + a_{22}c_2 + \dots + a_{2n}c_n)x_2 + \dots \\ &\quad (a_{m1}c_1 + a_{m2}c_2 + \dots + a_{mn}c_n)x_m \end{aligned}$$

β will be independent iff in the eqn

$$c_1 y_1 + c_2 y_2 + \dots + c_n y_n = 0$$

all c_i non zero

This is true if all the coefficients of x_i are zero but not all of c_i are zero

$\Rightarrow$ the homogeneous system of linear eqn in c_i

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

has a non-trivial soln

Since A is an $m \times n$ matrix with $m < n$

From Lemma 3.6 (1)

β is L.D.

Theorem (3.9) If a basis for a vector space V consists of n vectors, then so does every other basis.

Defn The dimension of a vector space V is the no. (say n), of vectors in a basis V ,

$$\dim V = n$$

If V has a basis of a finite number of vectors, V is said to be finite dimensional.

Ex

- (1) If V has only zero vector: $V = \{0\}$, then $\dim V = 0$
- (2) If $V = \mathbb{R}^n$, then $\dim \mathbb{R}^n = n$, since V has the standard basis.
- (3) If $V = P_n(\mathbb{R})$ of all polynomials of degree less than or equal to n , then $\dim P_n(\mathbb{R}) = n+1$, since $\{1, x, x^2, \dots, x^n\}$ is a basis for V .
- (4) If $V = M_{m \times n}(\mathbb{R})$ of all $m \times n$ matrices, then $\dim M_{m \times n}(\mathbb{R}) = mn$

Thm 3.10 Let V be a finite dimensional vector space.

- ① Any linearly independent set in V can be extended to a basis by adding more vectors if necessary.
- ② Any set of vectors that spans V can be reduced to a basis by discarding vectors if necessary.

$\Rightarrow$ A basis for a vector space V is a set of vectors in V which is maximally independent and minimally spanning.

Corollary-3.11 Let V be a vector space of dim n . Then

- ① any set of n vectors that spans V is a basis for V .
- ② any set of n linearly independent vectors is a basis for V .

Ex

Find a basis for $\mathbb{R}^4$ that contains the vectors $V_1 = (1, 0, 1, 0)$ and $V_2 = (-1, 1, -1, 0)$

Let $\{e_1, e_2, e_3, e_4\}$ are the standard basis for $\mathbb{R}^4$.

$$\text{Let } S = \{V_1, V_2, e_1, e_2, e_3, e_4\}$$

$\therefore \{e_1, e_2, e_3, e_4\}$ spans $\mathbb{R}^4$

$$c_1 V_1 + c_2 V_2 + c_3 e_1 + c_4 e_2 + c_5 e_3 + c_6 e_4 = 0$$

$$c_1 - c_2 + c_3 = 0$$

$$-c_2 + c_4 = 0$$

$$c_1 - c_2 + c_5 = 0$$

$$c_6 = 0$$

RREF

$$\left[\begin{array}{cccccc|c} V_1 & V_2 & e_1 & e_2 & e_3 & e_4 & \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

$\therefore V_1, V_2, e_1, e_4$ column have leading 1. $\Rightarrow \{V_1, V_2, e_1, e_4\}$ is a basis of $\mathbb{R}^4$.

Ex Let $S = \{V_1, V_2, V_3, V_4, V_5\}$ be a set of vectors in $\mathbb{R}^4$

$$V_1 = (1, 2, -2, 1), V_2 = (-3, 0, -4, 3), V_3 = (2, 1, 1, -1)$$

$$V_4 = (-3, 3, -9, 6) \text{ and } V_5 = (9, 3, 7, -6)$$

Find subset of S that is a basis for $W = \text{span } S$.

Solⁿ

$$C_1(1, 2, -2, 1) + C_2(-3, 0, -4, 3) + C_3(2, 1, 1, -1) \\ + C_4(-3, 3, -9, 6) + C_5(9, 3, 7, -6) = (0, 0, 0, 0)$$

$$C_1 - 3C_2 + 2C_3 - 3C_4 + 9C_5 = 0$$

$$2C_1 + C_3 + 3C_4 + 3C_5 = 0$$

$$-2C_1 - 4C_2 + C_3 - 9C_4 + 7C_5 = 0$$

$$C_1 + 3C_2 - C_3 + 6C_4 - 6C_5 = 0$$

RREF

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} & \frac{3}{2} & 1 & 0 \\ 0 & 1 & -\frac{1}{2} & \frac{3}{2} & -\frac{5}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$\therefore$ Columns 1 & 2 has leading 1.

$\Rightarrow \{V_1, V_2\}$ is a basis for $W = \text{span } S$.

Th^m

Let $S = \{V_1, V_2, \dots, V_n\}$ be a set of nonzero vectors in a vector space V and let $W = \text{span } S$.

Then some subset of S is a basis for W .